Adaptive Speculative Decoding

7-Minute Research Project Video Outline

General Format

- Format: slides with diagrams, tables, charts, and pre-recorded experimental results.
- Language: English.
- Style: calm, clear, and technical.
- Suggested number of speakers: 3–4.
- Main focus: the proposed method, experimental setup, and research results.
- All demonstrated code and results must match the frozen commit referenced in the final report.

0:00–0:35 — Title and Problem Statement

On Screen

Title slide:

Adaptive Speculative Decoding

Subtitle:

Dynamic draft length selection and online distillation for faster LLM inference

Team members:

- Andrey Polevoy
- Danil Popov
- Vadim Poponnikov
- Evgeniy Pestrovskiy
- Mikhail Krylov
- Aliya Khadeeva

Possible diagram:

User prompt → Large Language Model → Slow token-by-token generation

Narration

> Large language models are widely used in research, software development, data analysis, and many other applications. However, their autoregressive generation process can be slow because tokens are generated sequentially.

> Our project, Adaptive Speculative Decoding, explores how speculative decoding can accelerate this process. Our goal was to improve the generation speed of a target language model by using a much smaller draft model.

0:35–1:10 — Proposed Solution and Research Question

On Screen

Show a comparison between standard generation and speculative decoding.

Standard generation:

Target model → one token → one token → one token

Speculative decoding:

Draft model → proposes `k` tokens
Target model → verifies them in parallel

Highlight the main research question:

How should we choose `k` to maximize inference speed?

Narration

> In standard speculative decoding, a small draft model generates several candidate tokens. The larger target model then verifies these tokens in parallel.
>
> The number of proposed tokens is represented by `k`. If `k` is too small, we do not fully use the potential of parallel verification. If `k` is too large, the draft model may generate many tokens that are later rejected.
>
> Existing approaches commonly use a fixed value of `k`. Our main research question was whether selecting `k` dynamically could provide better performance.
>
> In parallel, we also explored online distillation, where information from the target model is used to improve the draft model during operation.

1:10–1:35 — Hypothesis, Audience, and Research Track

On Screen

Three blocks:

Hypothesis
Approximately 20% inference speedup

Target audience
Developers deploying LLMs locally

Research contribution
Dynamic selection of speculative decoding length

Narration

> Our main hypothesis was that adaptive speculative decoding could improve inference speed by approximately twenty percent compared with standard generation.
>
> The potential users of this approach are developers and researchers who deploy language models locally or operate under limited compute resources.
>
> Even a moderate improvement in inference speed can benefit every downstream task that depends on language model generation.

IMPORTANT: COMPLETE BEFORE THE FINAL RECORDING

Research Track Acceptance Criteria

Add a separate slide that explains:

- the official acceptance criteria for the Research track;
- how the project satisfies each criterion;
- which experiments, baseline comparisons, and results demonstrate that the criteria were met.

Possible narration:

> We selected the Research track. The project satisfies its acceptance criteria by formulating a clear research question, implementing and evaluating an adaptive method, comparing it with a baseline, and analyzing the resulting performance.

This wording must be checked against the official course requirements.

1:35–2:25 — Method and System Architecture

On Screen

Show the main pipeline:

1. The user provides a prompt.
2. The adaptive controller selects `k`.
3. The draft model generates `k` candidate tokens.
4. The target model verifies the candidate tokens.
5. Correct tokens are accepted.
6. Rejected tokens are corrected by the target model.
7. The target model outputs are used for online distillation.
8. The process repeats until generation is complete.

Models:

- Target model: **Qwen 2.5 7B**
- Draft model: **Pythia 70M**

Add a note:

VERIFY THE EXACT MODEL NAMES AND VERSIONS

Narration

> Our system consists of three main components: a target model, a smaller draft model, and an adaptive controller.
>
> We used Qwen 2.5 7B as the target model and Pythia 70M as the draft model. The draft model proposes a sequence of candidate tokens, while the target model verifies them.
>
> Based on the current generation state and previous acceptance behavior, the adaptive controller determines how many tokens the draft model should propose.
>
> When the draft model predicts the target tokens correctly, several tokens can be accepted after a single target-model verification step. This reduces the number of expensive target-model calls.
>
> We also experimented with online distillation by using signals from the target model to improve the smaller draft model.

Recommended Diagram Elements

Use different arrows or line styles to represent:

- the generation flow;

- the verification flow;
- the online-distillation feedback loop;
- the adaptive selection of `k`.

2:25–2:55 — Technology Stack and Experimental Setup

On Screen

Technology stack

- Python
- PyTorch
- Hugging Face Transformers
- Additional evaluation and data-processing libraries

Experimental setup

- Multiple repeated runs
- Comparison with standard target-model generation
- Average metrics reported across runs
- Offline execution without an internet connection

Narration

> The project was implemented in Python using PyTorch and the Hugging Face Transformers library.
>
> We compared our adaptive speculative decoding method with a baseline consisting of standard generation by the target model without speculative decoding.
>
> Each configuration was evaluated over multiple runs. We report average values because individual generation times can vary due to hardware conditions and differences between prompts.

2:55–3:25 — Dataset and Evaluation Protocol

On Screen

Title:

Datasets and Evaluation

Information to complete:

- Dataset: **[INSERT DATASET NAME]**
- Number of examples: **[INSERT DATASET SIZE]**
- Average prompt length: **[INSERT VALUE]**
- Generated sequence length: **[INSERT VALUE]**
- Number of runs: **[INSERT VALUE]**
- Hardware: **[INSERT GPU, CPU, AND MEMORY]**

Narration

> We evaluated the method using [DATASET NAME]. The dataset contained [NUMBER] examples.
>
> We used relatively short prompts and generated relatively short output sequences. No additional dataset preprocessing was applied.
>
> For each experiment, we measured generation speed, memory consumption, token acceptance rate, and the average number of correctly predicted draft tokens.

Important Slide Label

MUST BE COMPLETED BEFORE RECORDING

- exact dataset name;
- dataset size;
- number of experimental runs;
- generated sequence length;
- hardware specifications;
- generation parameters.

3:25–4:45 — Experimental Results

This should be the central part of the video.

On Screen

First, show a comparison table:

Method	Generation Speed	Speedup	Memory Usage	Acceptance Rate	Average Accepted Tokens
Standard target model	[VALUE]	1.00×	[VALUE]	N/A	N/A

Method	Generation Speed	Speedup	Memory Usage	Acceptance Rate	Average Accepted Tokens
Fixed-`k` speculative decoding	[VALUE]	[VALUE]	[VALUE]	[VALUE]	[VALUE]
Adaptive speculative decoding	[VALUE]	1.20×	[VALUE]	[VALUE]	[VALUE]

Replace every placeholder with the actual measured value.

Then show two or three graphs:

1. Generation speed by method
2. Speedup for different values or strategies of `k`
3. Acceptance rate and average number of accepted tokens
4. Memory consumption, if relevant

Narration

> The main result is that our adaptive method achieved approximately a twenty percent speed improvement compared with standard generation using the target model.
>
> The baseline represents generation without speculative decoding. Our method completed the same generation workload faster while preserving the output distribution of the target model.
>
> We also measured memory consumption, acceptance rate, the average number of accepted draft tokens, and overall generation throughput.
>
> The acceptance rate is important because speculative decoding is beneficial only when a sufficient proportion of draft tokens are accepted by the target model.
>
> A higher value of `k` is not always better. When the draft model predicts tokens incorrectly, generating and verifying additional candidates creates unnecessary computation. The adaptive mechanism attempts to balance these two effects.
>
> Because the reported values were averaged over multiple runs, the results are more representative than a single timing measurement.

Important Wording Note

Do not say:

> Our results are statistically significant.

Unless a proper statistical significance test was conducted.

Use one of the following instead:

> The improvement was consistent across repeated runs.

Or:

> We observed a stable average improvement across our experimental runs.

4:45–5:25 — Novelty and Interpretation

On Screen

Title:

What Did the Experiments Show?

Main findings:

- Dynamic `k` selection can outperform standard generation.
- The optimal draft length depends on model behavior.
- Acceptance rate alone does not fully determine end-to-end speed.

Separate highlight:

Main contribution: adaptive selection of draft length

Narration

> Our experiments support the original hypothesis: speculative decoding with an adaptive draft length can provide a meaningful inference speed improvement.
>
> The main contribution of our work is the dynamic selection of `k`. Instead of treating the draft length as a fixed global hyperparameter, we treat it as a value that can change during generation.
>
> The experiments also showed that not all seemingly promising strategies improve end-to-end performance. Some approaches improved acceptance-related metrics but introduced additional computational overhead.
>
> This is an important result because optimizing a proxy metric does not necessarily optimize actual generation speed.

Careful Novelty Wording

Avoid saying:

> Nobody has tried dynamic `k` selection before.

A safer formulation is:

> In the approaches we reviewed, draft length was commonly treated as a fixed parameter. Our project investigates an adaptive alternative.

This statement should be updated after completing the literature review.

5:25–6:00 — Challenges and Lessons Learned

On Screen

Challenges

- Many initial hypotheses did not improve real performance.
- Additional control logic could eliminate the speedup.
- Multiple research directions had to be narrowed down to one.
- Small models and short sequences limited the experiments.

Lessons learned

- Measure end-to-end latency, not only intermediate metrics.
- Test hypotheses independently whenever possible.
- Parallel research reduces blocking between team members.

Narration

> The main challenge was identifying a method that improved actual end-to-end performance rather than only improving intermediate metrics.
>
> We investigated several different directions, but many of the initial hypotheses did not work as expected. We eventually selected the most promising method and combined the successful components into one implementation.
>
> One technical lesson was that additional decision logic must be very lightweight. Even a theoretically useful adaptive strategy may fail to produce a real speedup if its own computation is too expensive.
>
> From a teamwork perspective, we learned that independent parallel investigation is effective for research projects. It allowed different hypotheses to be tested without blocking the entire team.

6:00–6:30 — Limitations and Future Work

On Screen

Two columns:

Limitations

- A very small draft model and a relatively small target model
- Small and relatively simple datasets
- Short generated sequences
- A limited range of model combinations
- No complete scaling-law analysis

Future work

- Test larger target and draft models
- Use longer contexts and output sequences
- Evaluate more model combinations
- Reduce the overhead of the adaptive controller
- Combine adaptive `k` with other successful hypotheses
- Study scaling laws
- Develop the project into a publication or production system

Narration

> Our study has several limitations. We used a 70-million-parameter draft model and a 7-billion-parameter target model, relatively small datasets, and short generated sequences.
>
> Therefore, the current results should not yet be generalized to every model size or workload.
>
> Future work should include experiments with larger models, longer sequences, and a wider range of datasets. We also want to reduce the overhead of the adaptive controller and combine this approach with several other promising hypotheses.
>
> With a broader experimental study, this work could potentially be developed into both a practical inference optimization and a research publication.

6:30–6:50 — Team Contributions

On Screen

Team Contributions

- All members: hypothesis generation, implementation, and experimental validation
- Andrey Polevoy: project coordination
- Aliya Khadeeva: reports and project documentation
- Final method: selected and integrated from the most successful experiments

Add more specific contributions for the other team members once they are defined.

Narration

> All team members participated in proposing hypotheses, implementing experiments, and evaluating different approaches.
>
> Andrey Polevoy coordinated the project and organized the overall workflow. Aliya Khadeeva was responsible for reports and project documentation.
>
> In the final stage, the team combined the strongest experimental results into the final adaptive speculative decoding method.

6:50–7:00 — Conclusion

On Screen

Large text:

Approximately 20% faster LLM generation

Below it:

Adaptive selection of `k` can make speculative decoding more efficient.

Narration

> To conclude, our experiments show that adaptive speculative decoding can accelerate language model generation by approximately twenty percent in our experimental setup.
>
> The project demonstrates that dynamic draft-length selection is a promising direction for more efficient LLM inference.

Suggested Speaker Distribution

Four-Speaker Version

Speaker 1

- Problem statement
- Research question
- Target audience
- Research track

Approximate time: **0:00–1:35**

Speaker 2

- Architecture
- Models
- Technology stack
- Experimental setup

Approximate time: **1:35–3:25**

Speaker 3

- Results
- Metrics
- Novelty
- Interpretation

Approximate time: **3:25–5:25**

Speaker 4

- Challenges
- Limitations
- Future work
- Team contributions
- Conclusion

Approximate time: **5:25–7:00**

This distribution reduces the number of voice transitions and makes the video feel more consistent.

Suggested Slide Deck

1. Title
2. Problem and motivation
3. Standard generation versus speculative decoding
4. Research question and hypothesis
5. Research Track acceptance criteria
6. System architecture
7. Models and technology stack
8. Dataset and evaluation setup
9. Main results table
10. Speed comparison graph
11. Acceptance-rate and memory results
12. Novelty and interpretation
13. Challenges and lessons learned
14. Limitations and future work
15. Team contributions
16. Conclusion

For a seven-minute video, some of these slides should be combined. A final deck of approximately **10–12 slides** would be appropriate.

Information That Must Be Added Before the Final Version

Critical Missing Information

- **RESEARCH TRACK ACCEPTANCE CRITERIA**
- **Exact dataset name or dataset names**
- **Dataset size**
- **Exact Qwen and Pythia model names and versions**
- **Hardware specifications**
- **Number of experimental runs**
- **Input and output sequence lengths**
- **Exact values for all metrics**
- **Fixed-`k` baseline results, if this baseline was evaluated**
- **Exact description of the dynamic `k` selection algorithm**
- **The role of online distillation in the final method**
- **Results with and without online distillation**
- **Frozen commit link or hash**
- **More detailed contributions from Danil, Vadim, Evgeniy, and Mikhail**

Recording Checklist

Before Recording

- Run all experiments in advance.
- Save the final tables and graphs.
- Confirm that all values match the final report.
- Verify the frozen commit.
- Make sure that all slide text is readable.
- Rehearse the full presentation with a timer.
- Prepare backup screenshots of every important result.
- Disable notifications before screen recording.

After Editing

- Watch the complete video using headphones.
- Check the volume level of every speaker.
- Make sure that all graphs and tables are readable.
- Check for frozen frames, incorrect slides, and abrupt cuts.
- Confirm that the audio and video are synchronized.
- Watch the video on another device.
- Open the uploaded video link in an incognito window.
- Upload the video before the submission deadline.